```
    d3.selectAll("circle").style("fill",function(p) {
      return p == d.source ? "blue" : p == d.target ? "green" : "lightgray";
    });
};
```

This nested if checks to see if a node is the source, which is set to blue, or if it's the target and set to green, or if it's neither and set to gray.

If you're interested in exploring arc diagrams further and want to use them for larger datasets, you'll also want to look into hive plots, which are arc diagrams arranged on spokes. We won't deal with hive plots in this book, but there's a plugin layout for hive plots that you can see at https://github.com/d3/d3-plugins/tree/master/hive. Both the adjacency matrix and arc diagram benefit from the control you have over sorting and placing the nodes, as well as the linear manner in which they're laid out. The next method for network visualization, which is our focus for the rest of the chapter, uses entirely different principles for determining how and where to place nodes and edges.

6.2 Force-directed layout

The force layout gets its name from the method by which it determines the most optimal graphical representation of a network. Like the word cloud and the Sankey diagram from chapter 5, the `force()` layout dynamically updates the positions of its elements to find the best fit. Unlike those layouts, it does it continuously in real time rather than as a preprocessing step before rendering. The principle behind a force layout is the interplay between three forces, shown in figure 6.7. These forces push nodes away from each other, attract connected nodes to each other, and keep nodes from flying out of sight.

In this section, you'll learn how force-directed layouts work, how to make them, and some general principles from network analysis that will help you better understand them. You'll also learn how to add and remove nodes and edges, as well as adjust the settings of the layout on the fly.

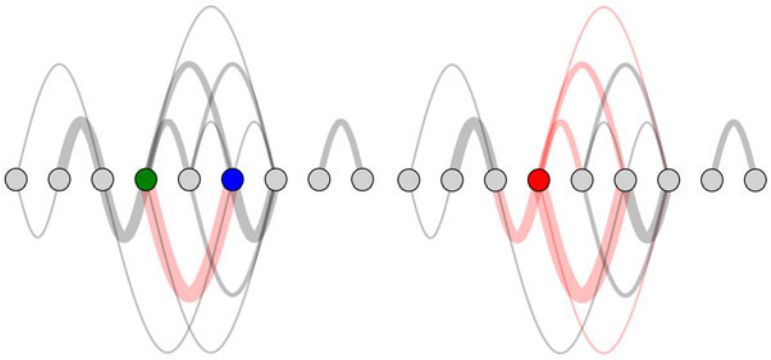


Figure 6.6 Mouseover behavior on edges (left), with the edge being moused over in pink, the source node in blue, and the target node in green. Mouseover behavior on nodes (right), with the node being moused over in red and the connected edges in pink.